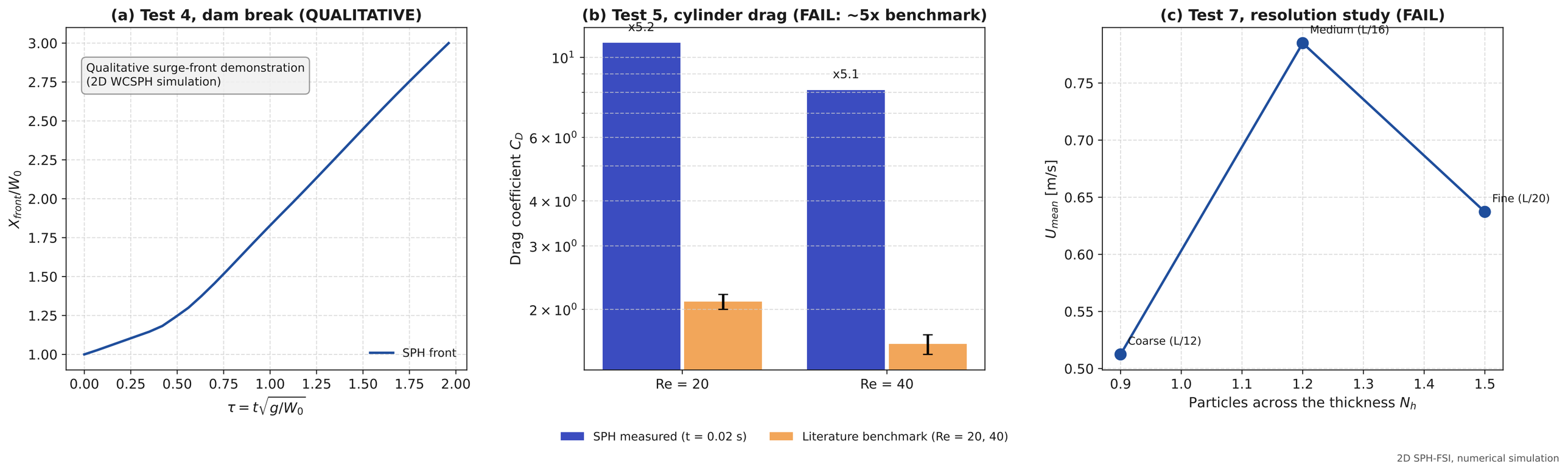
(a) Test 4, dam break (QUALITATIVE)
Qualitative surge-front demonstration
(2D WCSPH simulation)
SPH front
X_{front}/W_0
$\tau = t\sqrt{g/W_0}$
(b) Test 5, cylinder drag (FAIL: ~5x benchmark)
Drag coefficient C_D
x5.2
x5.1
Re = 20
Re = 40
SPH measured (t = 0.02 s)
Literature benchmark (Re = 20, 40)
(c) Test 7, resolution study (FAIL)
U_{mean} [m/s]
Medium (L/16)
Fine (L/20)
Coarse (L/12)
Particles across the thickness N_h
2D SPH-FSI, numerical simulation